



A modular agent-based framework for wildfire evacuation: Integrating fire spread, multimodal transport, and dynamic routing

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ARTICLE INFO

Keywords:

Wildland-urban interface
Wildfire
Evacuation
Simulation
Agent-based modeling

ABSTRACT

Wildfires increasingly threaten communities, especially in the Wildland–Urban Interface, where evacuation often becomes the only viable life-safety measure. Planning effective evacuations is challenging due to interactions among community behavior, road network capacity, and rapidly changing fire conditions. Despite advances in wildfire evacuation modeling, significant gaps remain: limited dynamic rerouting when roads are compromised, insufficient representation of multimodal evacuation, weak integration of high-resolution fire spread data, and the lack of inbound traffic and rescue operations. We introduce a modular, agent-based evacuation framework that combines fire spread, pedestrian movement, and vehicle traffic in a single model. It supports dynamic routing responsive to advancing fires, enables both outbound self-evacuation and inbound rescue operations, and incorporates multimodal transportation modes. A high-resolution wildfire model is coupled with diverse human behaviors and route availability. The model was tested in a case study in Portugal and validated against an evacuation drill, showing strong agreement in evacuation times and demonstrating its capacity to replicate real-world processes. Owing to its modular design and use of publicly available datasets, the framework can be readily adapted to diverse settings. It improves evacuation planning by allowing scenario testing, “what-if” analyses, and data-driven decision-making to enhance community safety and resilience.

1. Introduction

Wildfires have become an increasing threat to human settlements as climate change and land-use shifts intensify fire occurrence and severity (Liu et al., 2010; Bowman, 2020; Jolly, 2015). While fire is a natural ecological process, it becomes a societal hazard when it intersects with the wildland–urban interface (WUI), where residential development is directly exposed to flammable vegetation. Expansion of the WUI has substantially increased human exposure to wildfire hazards, with a growing proportion of fires affecting populated areas and infrastructures (Manzello et al., 2018; Radeloff, 2018). Climate-driven increases in temperature, drought, and fuel aridity accelerate fire spread and reduce suppression effectiveness (Liu et al., 2010; Jolly, 2015). At the same time, legacy fire-suppression practices have promoted fuel accumulation, contributing to larger and more severe events (USDA Forest

Service, 2003). Continued development in fire-prone landscapes, often coupled with flammable construction materials and limited road capacity, constrains shelter-in-place options and increases dependence on evacuation as the primary life-safety measure (Cova et al., 2013; Cova and Johnson, 2002; Dodson, 2010).

Recent wildfire disasters demonstrate that evacuation is usually the only viable strategy for protecting civilians. Large wildfire events in North America and Australia have resulted in mass evacuations involving tens of thousands of residents and substantial loss of life when evacuation processes failed or were delayed (Beverly and Bothwell, 2011; Teague et al., 2009; Wong et al., 2020a,b). Post-event analyses show that fatalities often occur during late, uncoordinated, or self-initiated evacuations, where individuals encounter fire on congested road networks or underestimate fire dynamics (Haynes et al., 2010; Rodrigues et al., 2022). These findings indicate that evacuation timing and

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<https://doi.org/10.1016/j.ssci.2026.107200>

Received 10 December 2025; Received in revised form 13 February 2026; Accepted 15 March 2026

Available online 21 March 2026

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coordination are as critical as the decision to evacuate itself.

Poorly managed evacuations can exacerbate risks. Simultaneous departures without traffic control or phased evacuation strategies can generate congestion, increasing exposure time and limiting access for emergency responders. Empirical evidence shows that early but unstructured evacuations may lead to higher casualty rates than well-planned, staged, and communicated strategies (Rodrigues et al., 2022). Effective evacuation, therefore, requires planning that explicitly accounts for fire progression, human behavior, and transportation constraints under rapidly evolving conditions.

In this context, predictive modeling and simulation have become essential tools for wildfire evacuation planning and emergency management. Simulation frameworks enable decision-makers to explore how fire spread, pedestrian movement, and vehicle traffic interact and to evaluate the effectiveness of alternative strategies, such as staged evacuation and public transport deployment (Ronchi et al., 2019; Gwynne et al., 2019; Intini et al., 2019). However, the operational value of these tools depends on the reliability of their assumptions and representations of both hazard and human response (Mitchell et al., 2023; Beyki et al., 2023). When appropriately applied, integrated evacuation models provide a systematic, evidence-based means of designing safer and more effective evacuation strategies for WUI communities facing escalating wildfire risk.

This work aims to develop an evacuation model that can support wildfire evacuation planning and management. The model integrates a high-resolution, wind-fire-coupled wildfire-propagation component with multimodal transportation and dynamic rerouting capabilities that account for road availability as it changes under the influence of advancing fire, as well as explicit modeling of pedestrian decision-making and movement, and of vehicle–pedestrian interaction. By combining these elements, the model seeks to provide a more realistic and adaptive framework for evaluating evacuation strategies and improving community safety.

The remainder of this paper is organized as follows: Chapter 2 reviews the state of the art in wildfire evacuation simulation, outlining key assumptions, limitations, and gaps in current knowledge, and then explains how this study seeks to bridge these gaps. Chapter 3 presents the methods and tools used to develop the proposed model. Chapter 4 reports the simulation results, while Chapter 5 interprets these findings in terms of their practical implications for decision-makers and stakeholders and outlines the study's limitations and directions for future research. Finally, Chapter 6 concludes the study by summarizing its key contributions and implications.

2. State of the art

Wildfire evacuation is a complex process shaped by fire dynamics, human decision-making, and the transportation system. Early studies primarily addressed road network capacity and clearance times under simplified assumptions about population response. Cova and Johnson (Cova and Johnson, 2002) showed that evacuation vulnerability is particularly acute in WUI communities with limited egress, where fire-induced road blockages can rapidly lead to entrapment. Wolshon and March (Wolshon and March, 2007) conducted an early subdivision-scale analysis of wildfire evacuation in the WUI using microscopic traffic simulation. Using CORSIM, they examined how evacuation clearance is constrained by the interaction between housing density, warning time, and egress capacity. Their results showed that short-notice evacuations and increased development density rapidly lead to queue formation and vehicle entrapment, while additional exits or longer lead times substantially improve clearance. This work provided early quantitative evidence that subdivision road design and evacuation timing are critical determinants of wildfire evacuation feasibility, anticipating later microsimulation and agent-based evacuation studies. Subsequent works coupled wildfire spread with transportation models, but typically relied on fixed participation rates, uniform vehicle access, and limited

behavioral realism (Dennison et al., 2007; Li et al., 2019).

Agent-based modeling (ABM) emerged to address these limitations by representing heterogeneous household behavior and adaptive decision-making. Beloglazov et al. (2016) presented an early coupled fire–traffic ABM, examining evacuation outcomes under different warning horizons. Although traffic congestion and fire-induced network loss were represented, evacuation participation was predefined, transport was limited to private vehicles, routing followed shortest paths to a single shelter, and fire spread was modeled deterministically. Results nevertheless demonstrated the dominant influence of warning timing on congestion, clearance, and entrapment risk. Similar assumptions appear in later studies. León and March (2017) analyzed bushfire evacuation in Victoria, Australia, assuming full compliance with evacuation orders and exclusive reliance on private vehicles. Varying warning times indicate that short-notice evacuations sharply reduce the likelihood of clearance and challenge the practical implementation of “shared responsibility” policies. Veeraswamy et al. (2018) focused instead on pedestrian evacuation under restricted road access, using the Exodus ABM platform. By holding population and fire progression constant and varying route availability, they isolated the effect of network loss on evacuation performance but excluded vehicular traffic and mixed-mode behavior.

A major advance was the development of integrated, multi-layer platforms that couple fire, pedestrian response, and traffic. Ronchi et al. (2017, 2019) emphasized the need for consistent spatial and temporal resolution across these layers. Building on this, Wahlqvist et al. (2021) introduced the WUI-NITY framework, integrating FARSITE fire spread, pedestrian response, and traffic evacuation within a modular Unity3D environment. The platform supports dynamic vulnerability mapping and evacuation trigger buffers via the PERIL algorithm, which compares available safe egress time with required egress time. Pedestrian response is represented macroscopically through departure-time distributions and network entry models, after which evacuees transition to vehicle-based traffic simulation. While WUI-NITY demonstrates the value of modular coupling, individual pedestrian movement, mixed-mode households, and emergency extraction are not explicitly modeled.

Case-specific ABMs have provided more detailed insights into household-level decision-making. Grajdura et al. (2022) modeled the 2018 Camp Fire using NetLogo, parameterizing agent behavior from post-event surveys. Routing was adaptive to fire-induced blockages, but the simplified, tree-like road network and reduced agent population amplified congestion and entrapment effects. Results highlighted the sensitivity of evacuation outcomes to awareness delays and vehicle availability. Similarly, Siam et al. (2022) developed a multimodal ABM for the 2018 Mati fire in Greece, coupling externally simulated fire spread with household decisions, pedestrian and vehicle flows, and shelter capacity. Their results showed that staged evacuations reduce congestion, shelter availability influences survival, and modal choice creates trade-offs between speed and exposure. However, fire avoidance during movement, car–pedestrian interactions, and mixed-mode evacuation at the individual level were not represented.

Overall, the literature shows a progression from macroscopic clearance-time analyses to integrated platforms and empirically informed ABMs. Persistent limitations remain. Fire progression is often imposed deterministically or externally, with limited dynamic interaction between fire and evacuees. Population response is commonly simplified through fixed participation rates or aggregate departure curves, omitting behaviors such as delayed response, shadow evacuation, or refusal. Evacuation is typically modeled as outbound movement to predefined shelters, with restricted representation of multimodal transport, emergency extraction, and car–pedestrian interactions. Traffic routing often relies on shortest-path algorithms, with simplified handling of fire-induced road loss. Pedestrian dynamics are often aggregated, and computational constraints limit population scaling. Addressing these gaps requires models that tightly couple fire dynamics with multimodal evacuation processes and dynamically adapt to evolving hazard and

network conditions. Table 1 presents a summary of the reviewed models, their features, and identified limitations.

This study aims to address gaps in the literature by developing an agent-based wildfire evacuation model that integrates several key advancements. The model incorporates a high-resolution, wind-coupled fire spread component, enabling a more realistic representation of hazard dynamics and their impact on evacuation processes. In this context, high resolution refers to fire spread simulations conducted at spatial and temporal scales capable of resolving fire–infrastructure interactions at the community level, rather than at the coarse regional scale. Such resolution is critical for evacuation analysis, as it enables more accurate estimation of time-to-impact on road segments, dynamic loss of egress options, and spatial variability in hazard exposure within settlements. However, higher resolution does not inherently imply

Table 1
Summary of the Evacuation Models.

Work	Features	Limitations
Cova and Johnson (2002)	– Road network capacity and clearance times; – focus on WUI communities with single exits	– Simplified assumptions; – no behavioral modeling; – static participation; – only the route system design emphasized
Dennison et al. (2007), Li et al. (2019)	– Coupled wildfire spread with transportation models	– Restrictive assumptions (fixed participation rates, uniform vehicles, limited behavior)
Beloglazov et al. (2016)	– Coupled fire-traffic ABM; – compared warning horizons; – private vehicles only; – shortest-path routing; – congestion modeled	– Fixed participation by warning type; – deterministic fire spread; – one shelter; – no multimodal transport
León and March (2017)	– Private vehicle evacuation; – nearest shelter; – varied warning horizons; – assumed full compliance	– All agents evacuate once warned; – no behavioral heterogeneity; – vehicle-only evacuation
Veeraswamy (2018)	– Exodus ABM; – pedestrian-only evacuation; – roads reserved for emergency vehicles; – single shelter	– No multimodality; – population and fire progression fixed; – no car-pedestrian interaction – outbound only
Wahlqvist (2021)	– Integrated fire (FARSITE), pedestrian response, and traffic; – modular framework; – trigger buffers; – dynamic vulnerability maps; – Destinations made unavailable as well as roads (latest update);	– pedestrians modeled macroscopically; – assumed car use after entry; – no mixed-mode households; – outbound only
Grajdura et al. (2022)	– ABM of Camp Fire in NetLogo; – survey-informed decision trees (awareness, departure, travel); – A* routing; – six shelters; – rerouting if fire blocks roads	– Scaled-down population (200 agents); – randomization instead of replication; – tree-like road network with no redundancy; – entrapment if rerouting fails – outbound only
Siam et al. (2022)	– Interdisciplinary multimodal ABM (NetLogo); – wildfire (FlamMap); – household decision-making; – staged vs simultaneous evacuation; – shelters and capacity modeled; – car and foot modes	– Outbound-only evacuation; – hazard treated as fatal if reached; – no avoidance; – no car-pedestrian interaction; – no mixed-mode households – outbound only

higher accuracy, as fire spread predictions remain sensitive to uncertain inputs and modeling assumptions. Consequently, recent works emphasize the value of high-resolution fire simulations when they are validated or corrected using observational data, such as satellite-derived active fire detections, and when their limitations are explicitly acknowledged in evacuation applications (Beyki et al., 2025a; 2025b). Unlike most existing frameworks that limit evacuation to outbound movement, the proposed model also accounts for bidirectional flows, thereby representing scenarios in which protection services conduct extraction operations into the hazard zone. The dynamic impact of fire on the transportation network is explicitly modeled, whereby fire blocks individual road segments rather than making entire destinations inaccessible, enabling more detailed interactions between fire spread and traffic flow. Pedestrian evacuation is modeled at the microscale, capturing detailed movement behaviors rather than relying on broad distributions. The model considers multiple transportation modes, i.e., on foot, private vehicle, and public transit, to better reflect real-world evacuation strategies. Pedestrian-vehicle interactions, including boarding and disembarking, are explicitly represented. In addition to self-evacuation, the framework incorporates the possibility of resident extraction by emergency forces, extending beyond individual households to include organized response operations. To demonstrate the proof of concept, the model is applied to a case study village at the wildland–urban interface, where local topography, demographics, and shelter locations are explicitly included. The results are assessed in terms of clearance times, the number of residents who reach safety, and the number at higher risk, providing insights into both scientific understanding and practical planning.

3. Methodology

3.1. Overview

The model simulates real-time interactions among dynamic fire behavior, transportation systems, and individual evacuation choices, allowing evacuation outcomes to emerge from the combined evolution of hazards and human responses. The framework integrates time-dependent fire-spread predictions with multimodal evacuation dynamics, including private vehicles, pedestrian movement, and assisted evacuation by emergency services. Fire arrival information is used to dynamically update road availability and routing constraints, while resident agents determine departure timing and transport mode based on evolving conditions. Traffic and pedestrian flows are represented at micro- and mesoscales, allowing congestion, rerouting, and mode switching to emerge during the simulation. Emergency services are modeled as active agents capable of entering hazardous zones, collecting evacuees, and transporting them to safe locations, enabling the concurrent simulation of self-evacuation and coordinated rescue operations. All model components, assumptions, and methodological contributions are described in detail and systematically evaluated in the following sections.

3.2. Agent-based modeling

The modeling environment adopted in this study is AnyLogic (Borshchev, 2013), a multi-method simulation platform that supports the integrated use of system dynamics, discrete-event simulation, and ABM. Built on JavaScript, models can be exported to be executed independently of the development environment on any Java-supported platform, facilitating portability and extensibility (Borshchev, 2013). Therefore, the model is platform-independent, and the source code will be publicly available. Given the need to represent heterogeneous evacuees, dynamic hazards, and adaptive interactions, an agent-based modeling approach was selected. Although built using AnyLogic as the simulation platform, the framework constitutes a fully developed modeling architecture in which agent definitions, statecharts, decision

rules, functions, and inter-module interactions are fully developed by the authors rather than predefined features.

Within the ABM framework, the system is represented as a collection of autonomous agents, such as residents, vehicles, and fire, each endowed with individual states, attributes, and behavioral rules. This bottom-up representation enables complex evacuation dynamics to emerge from local interactions rather than being imposed through aggregate assumptions (Borshchev, 2013; Grigoryev, 2024). Agents encapsulate both physical and decision-making components and may be static or mobile. Individual heterogeneity is represented through agent attributes and memory, defined via parameters (static characteristics) and variables (dynamic states), allowing differences in demographics, mobility, or risk perception to be explicitly captured (Grigoryev, 2024).

A combination of functions, events, and statecharts governs agent behavior. Functions implement reusable logics, such as route selection, risk assessment, or shelter choice, ensuring consistency across agents while maintaining modularity (Grigoryev, 2024). Events regulate periodic actions, such as hazard updates and re-routing checks, thereby synchronizing agent actions with evolving environmental conditions. Statecharts provide a formal structure for representing sequential and conditional behaviors, defining discrete behavioral states and transitions triggered by timeouts, conditions, messages, or spatial arrivals (Grigoryev, 2024; Harel, 1987). This structure is particularly suited to evacuation modeling, where actions such as preparation, movement, waiting, and boarding follow ordered yet adaptive sequences (Siam et al., 2022).

Agents operate autonomously while remaining responsive to interactions with other agents and infrastructures. Interactions may occur directly, through message exchange between agents, or indirectly via shared resources such as road networks, shelters, or queues. In AnyLogic, relations are represented through agent links, while global information and shared resources are managed at the environment level. Communication is supported through message passing and broadcast mechanisms, enabling both one-to-one and one-to-many exchanges and allowing coordinated responses to dynamic conditions (Grigoryev, 2024).

3.3. Wildfire behavior prediction model

Wildfire dynamics in this study were simulated using FireStation (Lopes et al., 2002), a software designed to predict fire spread across complex topography. FireStation integrates fire behavior with wind field simulations, and at its core, it applies Rothermel's surface fire spread model (Rothermel, 1972), which estimates the rate of spread and fire intensity as a function of fuel, slope, and wind.

A distinctive feature of FireStation is the modeling of wind fields using two approaches: NUATMOS and CANYON. NUATMOS, developed by Ross et al. (1988) is a linear model that solves the continuity equation on a three-dimensional grid. It interpolates observational wind data onto a grid and refines the data using variational analysis to produce a divergence-free flow field. Although it cannot account for buoyancy or separation effects, it yields realistic results at low computational cost, and its accuracy improves with additional meteorological data. CANYON, developed by Lopes et al. (1995) provides a more detailed representation by solving the three-dimensional Navier-Stokes equations with a turbulence closure using the $k-\epsilon$ model. It accounts for terrain and vegetation roughness as well as thermal feedback from the fire on the flow field. Although computationally more demanding, it yields more accurate wind fields under complex conditions. These coupled capabilities enable FireStation to simulate fire spread under dynamically evolving wind fields, thereby supporting realistic integration with evacuation modeling.

The required inputs for running FireStation include a digital elevation model to represent topography, a fuel map describing vegetation types and fuel model to present their associated fire behavior parameters, and weather conditions, primarily wind speed and direction. The

main outputs are the rate of spread, the evolving fire perimeter, fireline intensity, flame length, and the time of ignition for each location. Within the software, these outputs are visualized through a graphical interface that displays the spatio-temporal progression of the fire. To couple these results with the evacuation model, the outputs are exported to.csv format, providing the calculated fire characteristics and corresponding coordinates for each cell of the simulation domain. This table is then imported into AnyLogic, where it is used to initiate fire agents and to dynamically model fire propagation in space and time, ensuring the hazard evolution.

3.4. Evacuation model

The evacuation model is constructed as a set of interconnected components that collectively represent the complexity of evacuation. First, the area of interest (AOI) must be defined, as it determines the simulation's spatial boundaries and provides the geographic context for hazard spread and population movement. Within this area, agents interact, each guided by its own behavioral rules, features, and constraints. The developed model consists of six agents: pedestrians, personal vehicles, extraction vehicles, routes, waypoints, and fire, along with a hierarchical agent that facilitates interaction among all agents. Together, these components constitute the key elements of the evacuation system in response to a wildfire threat, enabling coherent simulation of their interactions.

An agent is created as the central environment, which acts as the supervisory layer of the model. This environment contains the global structures and data registries that coordinate the activities of the other agents. For instance, it holds the spatial representation of the road network, manages time-dependent information on fire-related closures, and maintains the registry of waypoints that vehicles must follow during their trips. It also provides the space in which agents interact. This environment serves as the integrator that connects all subsystems, ensuring that agents' behaviors are synchronized. The remaining agents are embedded within the central environment and interact with one another according to their specific behavioral rules and environmental constraints. Their structure and function are detailed in the following sections.

3.4.1. Area of interest

To define the AOI, the model uses GIS map functionality integrated into the main simulation environment. The base map is obtained directly from OpenStreetMap (OSM) servers, allowing the user to search for and select a specific location or region by name. Once the AOI is established, it is necessary to specify the spatial distribution of key elements within the community, including residential buildings; designated shelters, either within the community for shelter-in-place strategies or at safer external locations for refuge; and emergency services facilities, such as civil protection or firefighting stations. When available, these data can be imported directly in shapefile format, ensuring accurate georeferenced placement of features. In the absence of such datasets, the required elements can be defined manually. This configuration provides the initial spatial distribution of agents, including residents, vehicles, and extraction units.

Rather than relying on the standard OSM road network, the model uses road network data in.pbf file format specific to the study area. These files provide detailed road infrastructure data and can be extracted at flexible resolutions, ranging from coarse representations that include only major roads to highly detailed networks that also cover small passageways, emergency routes, and pedestrian or hiking paths. They are freely accessible online and can be obtained for virtually any region. Based on the selected road network and the locations of importance, the model generates two additional agent types: routes and waypoints. Each route is defined by three parameters: its origin, destination, the link representing the road segment between them, and a variable that is its availability status. Waypoint agents, which represent

intermediate nodes between an origin and a destination, are defined by three parameters: the origin, the destination, and the sequence of intermediary locations that connect them. The rationale for using PBF data instead of OSM, as well as the advantages of defining routes and waypoints as agents, is discussed further in the navigation section.

3.4.2. Fire

The fire is represented as a list of ignited points, each defined by its geographic coordinates (x_f) and the corresponding time of ignition (τ_f). In principle, fire agents can be generated individually at the exact moment their ignition time occurs in the simulation. However, this approach increases computational runtime, as the model must repeatedly access and process the fire dataset at each time step. To improve efficiency, fire points can be grouped into temporal intervals (Δ) so that all points within a given time range are activated simultaneously. The current time bucket is calculated as $k(t) = \lfloor t/\Delta \rfloor$. All fires are grouped into these buckets based on their ignition times, forming sets $\mathbb{F}_k$:

$$\mathbb{F}_k = \{f \in \mathbb{F} : \lfloor \tau_f/\Delta \rfloor = k\}$$

At the start of each bucket, all fires belonging to that group become active according to:

$$\alpha_f(t^+) = \begin{cases} 1, f \in \mathbb{F}_{k(t)} \\ \alpha_f(t^-), \text{Otherwise} \end{cases}$$

here, $\alpha_f(t)$ indicates whether a fire f is active (1) or inactive (0).

For example, by applying a 30-minute bucketing scheme, all fires with ignition times between 0 and 29.999 min are activated simultaneously, followed by those between 30 and 59.999 min, and so on, thereby forming 30-minute burning isochrones. In this way, the model reduces the number of database queries while maintaining temporal accuracy in the representation of fire spread. This procedure is implemented through a periodic event that triggers every 30 min and continues for as long as active fire points remain. Note that these values are illustrative and can be defined by the user based on the desired level of granularity.

3.4.3. Fire and route interaction

To evaluate the effect of fire on the road network, a function is defined to check the proximity of active fire points to the routes. This function relies on the grouped fire points described earlier and calculates the Euclidean distance between each fire in a given time bucket and all route segments ($e \in E$). If a fire point is found to be closer than a specified threshold (d^*), the corresponding route is flagged as unavailable and added to a list of blocked segments.

$$S_k = \left\{ e \in E : \min_{x \in e, f \in \mathbb{F}_k} d(x_f, x) \leq d^* \right\}$$

here, S_k is the set of roads that will be closed in time bucket k . Once the simulation reaches that time, the availability of those roads (a_e) changes according to:

$$\alpha_e(t^+) = \begin{cases} 0, e \in S_{k(t)} \\ \alpha_e(t^-), \text{Otherwise} \end{cases}$$

The variable $a_e(t)$ represents the availability of a road segment (1 = open, 0 = closed).

The same procedure is repeated for each successive set of fire buckets as the simulation advances and the fire propagates through the landscape. The model performs real-time checks to close the roads that the fire gets too close to them. At time t , let $\mathcal{A}(t)$ denote the set of all active fires. A road is closed if its Euclidean distance along the route to any active fire is less than or equal to d^* :

$$\text{If } \alpha_e(t^-) = 1 \text{ and } \min_{x \in e, f \in \mathcal{A}(t)} D_G(x_f, x) \leq d^* \implies \alpha_e(t^+) = 0$$

Here, $D_G(x_f, x)$ represents the distance along the road network between the fire's location and the road.

In this way, the function progressively updates road availability by evaluating the proximity of each new fire bucket to the network. This function is executed periodically at user-defined time intervals and dynamically updates the road availability throughout the simulation. Importantly, if a route has already been flagged as unavailable in a previous check, it is not re-evaluated in later iterations. This prevents redundant calculations and reduces runtime.

3.4.4. Pedestrians

To represent pedestrian behavior in the evacuation process, a dedicated agent population (type) is created to model the residents of the community under analysis. The characteristics of residents are defined through a set of parameters that capture individual differences in evacuation behavior. These parameters include the walking speed, the time required to prepare before leaving, the probability of choosing to evacuate or remain in place, the probability of self-evacuation, and whether they have a personal vehicle. These values are assigned uniquely to each resident, allowing the model to reflect heterogeneity within the population.

Each resident is treated as an autonomous agent capable of exhibiting individual behavior, allowing the model to capture heterogeneity in responses. The dynamics of this behavior are formalized through a statechart that defines the sequence of actions and decisions taken by the resident. The actions undertaken by the resident depend on the state they occupy in the statechart, with each state linked to specific coded instructions that are executed automatically upon entry. At the evacuation onset, agents may choose either to stay or to evacuate. Among those who evacuate, some residents initiate self-evacuation once the advancing fire front reaches a user-defined proximity threshold. This allows the simulation to model spontaneous departures triggered by perceived danger as the fire approaches, and the remainder to initiate evacuation following a mandatory evacuation order that the user can issue at any time. For all residents who decide to leave, a preparation phase follows, during which they perform household-level tasks before departure. The duration of this phase is modeled stochastically within a user-defined time range. Once preparation is complete, residents proceed on foot at walking speeds, which vary with individual physical ability, to a designated meeting point, where they remain until rescue vehicles arrive. Boarding introduces an additional delay, defined by a user-specified time range, that represents the process of entering and securing a place in the vehicle. After boarding, the resident is transported by vehicle to a designated refuge located in a safe area outside the hazard zone.

The number of residents (community population) can be generated from a data table that specifies the number of occupants in each building. This allows for a spatially accurate initialization of the population across the area of interest. In the absence of such data, population values can be assigned manually or generated by randomly distributing residents across buildings while ensuring that the total matches the expected population of the community. This representation ensures that decision-making, preparation, walking, waiting, boarding, and transportation are all explicitly governed by the resident's current state and its associated actions.

3.4.5. Vehicles

To model vehicle behavior during evacuation, two agent types are created to represent personal vehicles and emergency extraction vehicles. The characteristics of vehicles are represented by a set of parameters that determine their operational behavior during the evacuation. These include the time required before dispatch (for emergency vehicles), which reflects delays prior to departure; the maximum passenger capacity, which limits how many residents can be transported at once; and the maximum attainable speed, which depends on the type and

condition of the road segment being traversed. Each vehicle can be assigned distinct values for these parameters, ensuring variability in performance and enabling the model to capture differences in transport availability and efficiency across the evacuation system. Vehicle movement is represented using a mesoscopic network-based approach. Each vehicle is modeled as an individual agent that follows a dynamically updated route through the road network. Travel times are assigned at the link level and remain fixed during the simulation; they are not influenced by the number of vehicles on the same link.

These agents operate within the road network using a routing algorithm that guides them toward their destinations, which may consist of multiple points connected through intermediate destinations or midway points. As with residents, vehicle agents are governed by statecharts that define their behaviors and decisions, with each state executing a specific action during the evacuation process. Emergency vehicles are initially positioned at civil protection or emergency service bases, and personal vehicles are positioned at the WUI community. Upon receiving an evacuation order, emergency vehicles transition from an idle or standby state to active deployment, following dynamically updated routes toward the endangered community. Once there, they collect residents at designated meeting points and transport them to safe locations outside the hazard zone. For personal vehicles, they don't use the customized routing function as they are assumed to be oblivious to the hazard development and may use roads threatened by the growing fire. As they follow their chosen route, their minimum proximity to the fire is measured to determine their risk level during the evacuation process. This representation allows vehicle agents to capture both the logistical aspects of evacuation, such as routing and rerouting, and the operational dynamics of emergency services tasked with extracting and relocating populations under wildfire threat, as well as risks associated with unorganized evacuation.

3.4.6. Pedestrian and vehicle interaction

The boarding and disembarking of residents are managed through a cyclic event that coordinates both the loading process and the departure of buses. Two conditions define residents waiting for evacuation: they must be at a meeting point ($m_i = 1$) and not yet boarded ($q_i = 0$). The set of waiting residents is:

$$W(t) = \{i \in \mathcal{R} : m_i(t) = 1, q_i(t) = 0\}$$

When passengers are waiting at a meeting point, the event triggers a function that dispatches buses and assigns their destinations. Upon arrival, the bus sends a message indicating its presence, which is received by the waiting residents. The bus then transitions from a driving state to a loading state, during which residents begin boarding. The boarding process is iterative: the model checks the bus's available capacity, and for each resident who boards, one is removed from the waiting list and added to the vehicle's passenger load. When a bus is loading and has free capacity, a subset of waiting residents, $A_b(t)$ boards the vehicle. The number of passengers on the bus and the residents' boarding status are updated as follows:

$$l_b(t^+) = l_b(t) + |A_b(t)|; q_i(t^+) = 1 \text{ for all } i \in A_b(t)$$

Here, $l_b(t)$ is the number of passengers currently on the bus.

This continues until either the bus reaches full capacity or no additional passengers remain. Each resident who boards transitions to an in-vehicle state, and once the bus is full, it receives a "loaded" message and switches to the dispatch state, initiating movement toward safety. At this point, the bus applies routing logic to reach its assigned destination. In addition, a supporting function continuously updates the number of residents still waiting. If this number is greater than zero, the function either directs the same bus to return or assigns another vehicle (if available) to complete the boarding process.

3.4.7. Routes and navigation

In most evacuation models, routing is handled through shortest-path or fastest-path algorithms commonly used in routing services, such as A* or Dijkstra (Grajdura, 2022; Siam et al., 2022). While effective in standard applications, these approaches present limitations when applied to wildfire evacuation. Specifically, disabling individual road segments as they become compromised by fire is difficult with these algorithms, as the only practical method is to render the entire destination unreachable (Wahlqvist, 2021). This is unsuitable for the present study, as evacuation traffic flow is not solely outbound, and destinations are not limited to shelters; rather, the endangered community itself can serve as a destination for emergency services extraction operations. Furthermore, the routing framework provided by OSM does not support the dynamic removal of individual road segments during simulation, because road elements are instantiated before execution and cannot be deleted at runtime (Borshchev, 2013; Grigoryev, 2024). To overcome these challenges, a custom routing logic was developed that dynamically marks roads as available or unavailable during the simulation. This approach ensures that routing decisions adapt to the evolving hazard conditions while maintaining the flexibility required to represent both outbound evacuations and inbound rescue operations.

Two functions were developed to provide navigation logic for vehicles, ensuring that they avoid roads located within the hazard zone while still reaching their designated destinations.

The first function begins by checking whether a direct route exists between the vehicle's current location (o) and its intended destination (d), and whether this route remains available, meaning it has not been compromised by the fire. If the direct route is available ($a_{(o,d)} = 1$), the vehicle proceeds along it.

$$a_{(o,d)}(t) = 1 \implies \text{choosepath}P = [o, d]$$

If not, the function consults the waypoint record, which contains chains of intermediate points ($v_1, v_2, \dots, v_m$) linking the same origin and destination. Each segment along the route must be open for the vehicle to use that alternative path:

$$\begin{aligned} \text{selectfirstchain}(v_1, v_2, \dots, v_m) \text{ where all legs are open} &\longrightarrow P_b(t) \\ &= [o, v_1, v_2, \dots, v_m, d] \end{aligned}$$

Here, $P_b(t)$ is the ordered list of intermediate points (waypoints) that define the route for bus b .

Each chain represents an ordered sequence of nodes that can serve as alternative routes, expressed as a series of legs from the starting point through midpoints (Origin $\rightarrow$ midpoint1 $\rightarrow$ midpoint2 $\rightarrow \dots \rightarrow$ Destination). The function evaluates these chains in order and selects the first chain for which all legs are available. Once a valid chain is found, the vehicle is directed to start moving along this path, and its upcoming waypoints are registered. This waypoint record is vital, as it serves as the basis for the second function, which advances the vehicle step by step along its assigned chain until it reaches the final destination. In this way, vehicles adaptively navigate around blocked segments while maintaining the continuity of evacuation operations.

The second function advances vehicles step by step along their assigned chain of midpoints whenever a vehicle is idle. At each iteration, the function compares the vehicle's current location with the next midpoint in its route list. If the vehicle has not yet reached that midpoint, it is instructed to move there. Once the vehicle reaches the final waypoint in the sequence, it sends an "arrived" message, signaling the start of the loading or unloading process for residents.

The function stores the remaining route to the destination as an ordered list of midpoints and continuously checks the vehicle's progress along this list. If the vehicle's current location equals the next midpoint, the function determines whether additional midpoints remain. If no further nodes exist, the vehicle is marked as having arrived; otherwise, it advances to the next node in sequence. To improve efficiency, the function also creates a temporary list of vehicles that have completed

During execution, the function iterates over all vehicles but acts only on those with an assigned route list and that are not currently in motion. Vehicles already in motion are left unaffected and will be rechecked in subsequent iterations once they stop or reach their destination. This ensures that the stepwise advancement logic is applied consistently without interrupting vehicles already in transit.

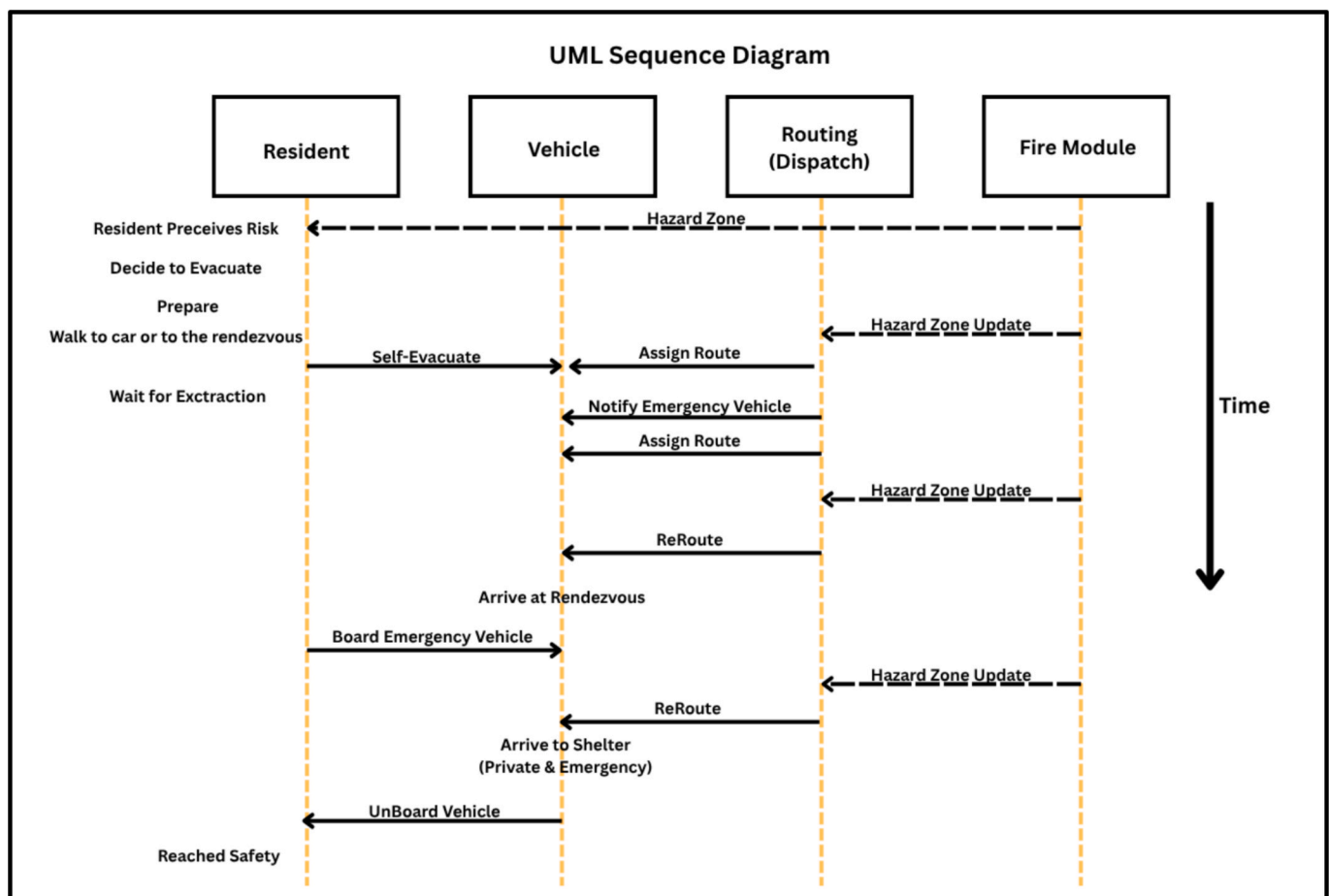
Fig. 1 presents a Unified Modeling Language (UML) sequence diagram illustrating the main interactions among the core components of the evacuation model: residents, vehicles, routing (dispatch), and the fire component. UML is a standardized graphical notation for describing the structure and dynamic behavior of complex systems. In simulation modeling, UML diagrams are used to visualize the sequence of interactions between components and the flow of information through time.

The model is structured upon five main components: fire, routing

Fig. 2 presents the conceptual framework and overall workflow of the model, illustrating the sequence of processes from model setup to output generation. The setup phase defines the spatial and environmental inputs, the fire behavior processed in FireStation, community data, and the road network imported into the AnyLogic environment. During the simulation phase, the system operates through interacting agents. Each agent type follows a defined behavioral logic represented through statecharts and decision points.

4.1. Case study

The case study selected to demonstrate the capabilities of the presented model is the village of Cerdeira, located in the Serra da Lousã, Coimbra District, central Portugal. This small settlement lies in a steep valley at an altitude of approximately 700 m. It is characterized by narrow access roads, dense vegetation, and a predominance of tourism-related activities. This socio-demographic profile of Cerdeira,



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Table 2
Summary of inputs and outputs of the model components.

Component	Inputs	Outputs
Fire	–Topography –Fuel map and model –Meteorological data –Ignition location and time	–Rate of spread –Flame length –Fireline intensity –The spatial and temporal evolution of the burning area
Routing and AOI	–Road network; –Coordinates of origins, destinations, meeting points, shelters, and safe zones; –The spatial and temporal evolution of the burning area –Building locations –Number of residents per dwelling	–Route and waypoint agents –Route assignments for vehicles –Updated road availability –Rerouting decisions
Resident	–Departure time distribution –Probability of evacuation –Walking speed –Preparation time –Car ownership	–Departure time –Mode of evacuation (on foot or by vehicle); –Arrival time at meeting points or shelters –Waiting and boarding times –Evacuation completion status
Vehicle	–Number and type of vehicles –Capacity –Speed;	–Vehicle trajectories; –Boarding and unloading events; –Travel times; –Arrival time at shelters; –Evacuation timelines
Integration	–All outputs from above	–Number of trapped or endangered residents

characterized by a transient tourist population and an elderly, low-density resident base, is representative of many fire-prone WUIs.

Field inspections began in November 2022, with on-site assessments of housing typologies, demographic characteristics, vulnerable inhabitants, and road conditions. Given the settlement's touristic character, an additional survey was conducted to inventory local accommodations and to understand visitor dynamics throughout the year. These efforts facilitated direct engagement among the research team, the local government, and the community, enabling the design and testing of protection measures within the EVACUARFLORESTA project framework (Santiago and Laím, 2025).

The evacuation exercise, held on 15 March 2023, simulated a fast-moving wildfire approaching from the southwest valley with the potential to reach the nearby São Lourenço chapel (rendezvous point) and threaten the village perimeter. The scenario assumed around 25 tourists and 5 residents in the area. Participants were instructed to gather at the designated refuge point located in the village square (shown as the rendezvous point in Fig. 3) before an organized evacuation by road was executed towards the Lousã Fire Station (can be seen in Fig. 4). The entire operation was coordinated by the project's research team in collaboration with the Lousã and Serpins Fire Brigades, National Republican Guard (GNR), and Municipality of Lousã.

The exercise populated the village to near full occupancy (30 individuals) using volunteers who acted as tourists, supported by multiple observers strategically positioned across the site. Evacuation began at 10:35 a.m., following a briefing at the Lousã Fire Station, and concluded at 12:15 p.m., when all vehicles had arrived at the designated safe zone. The time log shows that evacuation from alert to the refuge took approximately 22 min, and the total evacuation duration from initiation to arrival at the fire station was 1 h and 35 min. The exercise successfully tested communication, coordination, and the capacity of narrow road networks to support organized evacuation under pressure. The evacuation times reported by the evacuation conductors are presented in Table 3. The exercise did not include self-evacuation by private car due to ethical considerations and government restrictions on safety

exercises. Evacuation was only done through GNR as a convoy of cars, escorted by emergency vehicles (as an extraction). However, the simulated case includes self-evacuation to better reflect a more realistic situation.

In the model implementation, the geographic layout of Cerdeira was recreated using the GIS feature, with base layers obtained from OSM. The coordinates of residential buildings and roads were imported from geospatial data (.pbf file for roads and .shp file for buildings) and manually adjusted to match the village's actual layout. Each building was assigned a set of residents based on the observed demographic composition during the drill. The location of the local shelter used in the evacuation drill was specified within the model as the designated refuge point, and the surrounding road network was incorporated. This setup ensured that the simulation environment accurately represented the community's physical and social structures. Fig. 4 presents the AOI for the simulation in the software platform. The points and routes of interest for the simulation are marked on the GIS map layer.

The simulation parameters were defined to represent the evacuation drill scenario with a heterogeneous population and behavior. A total of 30 resident agents were initialized, each characterized by distinct mobility, preparation, and evacuation decision attributes. Walking speed followed a normal distribution with a mean of 1.1 m/s, bounded between 0.5 m/s and 1.7 m/s, reflecting documented typical pedestrian movement variability under emergency conditions, in hilled areas, and for people with less physical abilities (Giannoulaki and Christoforou, 2024). Preparation time was modeled as a normally distributed variable with a mean of 20 min, ranging between 15 and 30 min, and a standard deviation of 5 min based on field studies done from prior evacuations in Portugal (Santiago and Laím, 2025). Each agent was also assigned a probability of owning a private vehicle, where 20% of residents were considered to have access to a car. The decision to evacuate was modeled as a Boolean function with a probability greater than 80%, representing the subset of agents who decide to evacuate. The risk perception self-evacuation distance was set to 2000 m. Bus transport was available, with a maximum capacity of 40 passengers per vehicle, and fire progression was assessed at 30-minute intervals (time buckets).

4.2. Simulation results

The simulation ran in 71.37 s on a PC equipped with a 3.4 GHz processor and 32 GB of RAM, corresponding to a duration of 300 min. Of 30 residents, 25 (83.3%) chose to evacuate, while 5 (16.7%) remained. Among evacuees, 8 of them (32%) engaged in self-evacuation, triggered by proximity to the advancing fire, whereas 17 (68%) were extracted. Table 4 presents the total number of residents, the evacuation ratio, and the self-evacuation/extraction ratio.

Self-evacuated residents, activated by immediate fire proximity, showed shorter overall waiting and travel durations, reflecting direct, pedestrian movement to their cars and then to the shelter. Their mean total time to safety was 11.35 min.

Extracted agents, in contrast, were subject to formal coordination and vehicle staging, resulting in longer overall clearance times. Although they arrived at the rendezvous point at similar times (3.33 min), their mean waiting time was 30.61 min, largely due to delays associated with waiting for the emergency vehicle, dispatch, travel to the village, assembly, and boarding before departure. The uniform 44-minute travel time to safety corresponds to the collective bus evacuation phase. Table 5 presents the mean

Fig. 5 illustrates the cumulative arrival of residents at safety over time. The x-axis indicates the time at which each agent reaches safety (in minutes, measured from the first risk perception or the start of the evacuation), while the y-axis shows the cumulative number of evacuees who have completed their evacuation. The plot reveals a two-phase evacuation pattern: an initial period of gradual arrivals between 30 and 45 min, corresponding to self-evacuated residents who reached the safety zone independently, followed by a long plateau during which no

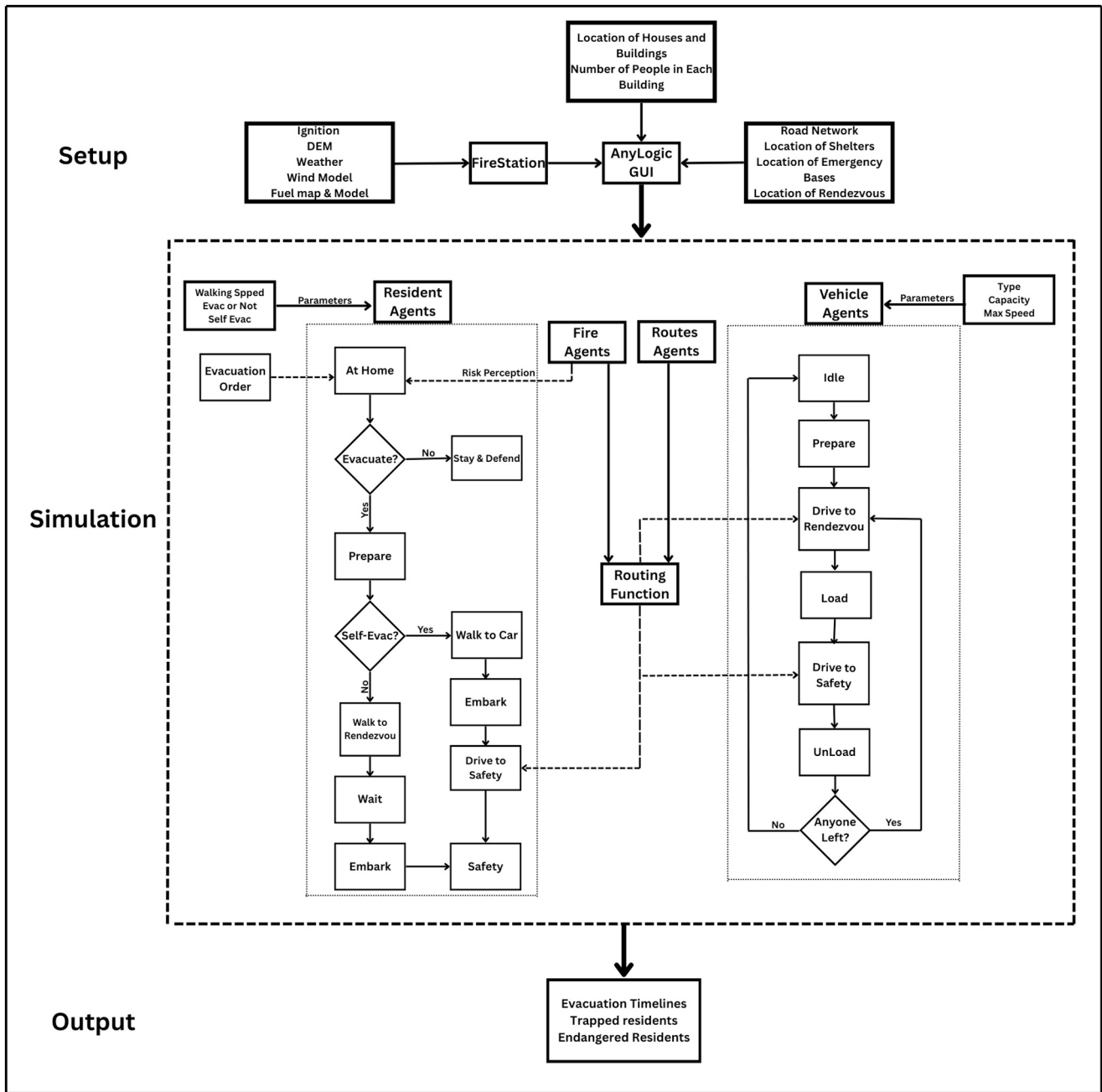


Fig. 2. Conceptual framework and workflow of the wildfire evacuation simulation model.

additional arrivals occur. The second phase begins around 100 min, when the remaining evacuees, those extracted, arrive collectively at the safety area.

Because self-evacuations occur spontaneously and are not coordinated by authorities, self-evacuating residents do not follow the routing logic implemented in the model to avoid endangered areas. Consequently, their chosen paths may intersect roads or zones that are compromised by the fire. To quantify hazard exposure, the model records the minimum distance between each self-evacuated agent and the fire agents at each time step. This metric indicates the extent to which evacuees approach hazardous zones during their journey. Existing frameworks, such as WUI-NITY (Wahlqvist, 2021), assess evacuation safety primarily through a time-based feasibility lens, comparing Available Safe Egress Time (WASET) and Required Safe Egress Time (WRSET) to determine whether evacuation can be completed in time at the community level. In contrast, the methodology proposed here

evaluates safety from a trajectory-based exposure perspective, quantifying the proximity of evacuees and the evolving fire front along the evacuation. The approaches are complementary: WUI-NITY addresses situations in which evacuation becomes infeasible, whereas the proposed method assesses the safety of evacuation under dynamic wildfire conditions.

The results from this metric are presented in Table 6, which indicates a clear pattern: as residents evacuate later, the advancing fire front progressively reduces the buffer of safe distance available to them, leading to smaller minimum distances between their vehicles and the active fire perimeter during their travel to the shelter. This reflects the consequence of fire propagation over time and demonstrates how delaying departure increases potential exposure to hazardous zones. Despite departing significantly later than all self-evacuated residents, the extraction vehicle recorded the largest minimum distance to the fire. This outcome occurs because the extraction vehicle was guided by the

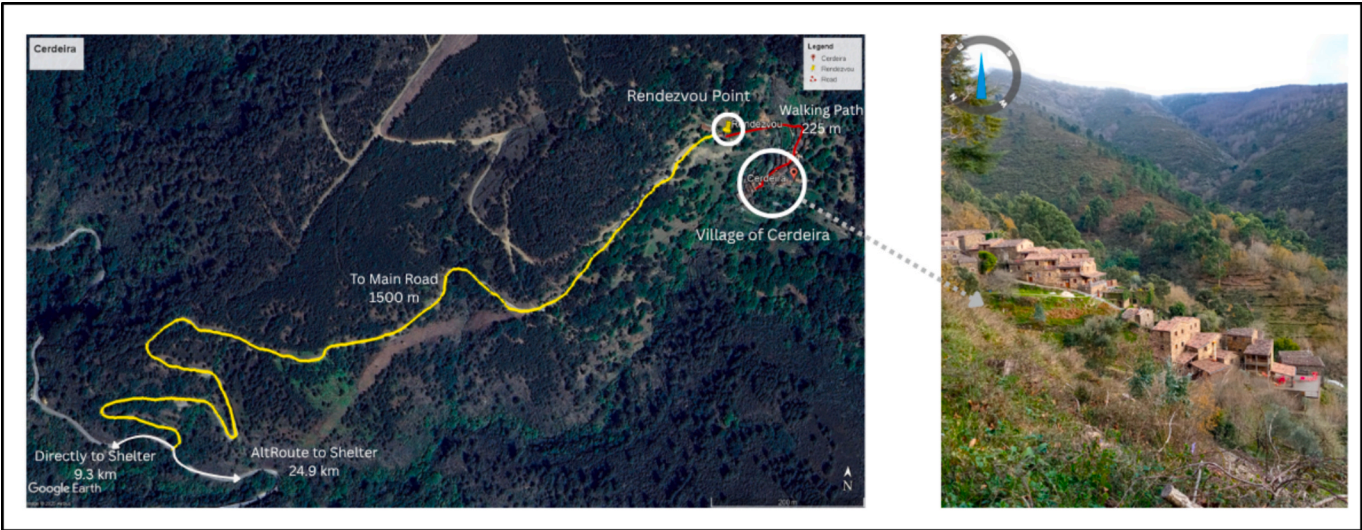


Fig. 3. Aerial photos and a map of Cerdeira.



Fig. 4. Simulation AOI.

model’s routing system, which actively avoids roads threatened or already compromised by the fire, thereby ensuring a safer trajectory even when evacuation begins at a later stage.

The extraction phase of the simulation was supported by a dedicated bus convoy that transported evacuees from the rendezvous point in Cerdeira to the designated shelter. The operational performance of the bus was evaluated using four temporal indicators: dispatch time, travel

time to the village, loading time, and travel time to safety. Table 7 shows the extraction vehicle’s performance.

Fig. 6 illustrates the simulation sequence. As the flames approach the settlement, residents with a higher perceived risk of harm initiate self-evacuation using their private vehicles once the fire front is within 2 km of their location. When the wildfire continues to propagate toward the village, the authorities issue a formal evacuation order, which can be

Table 3

Evacuation exercise times.

From alert to all occupants arrive at the rendezvous (preparation + walking)	22 min
Wait time at the rendezvous until extraction from the village	18 min
Travel time from the village to the shelter (with Vehicle, through the alternative route)	50 min

Table 4

Count and percentage of evacuated residents simulated.

Category	Count	Percentage
Total agents	30	100%
Evacuated	25	83.3%
Self-evacuated	8	32.0% of evacuees
Extracted (by bus)	17	68.0% of evacuees
Did not evacuate	5	16.7%

Table 5

Residents mean performance indicators.

Parameter	Self-evacuation	Extraction
Walking speed (m/s)	1.02	1.17
Preparation time (min)	21.8	20.56
Time to Rendezvous point (min)	4	3.33
Waiting time (min)	0.93	30.61
Time to shelter (min)	11.3	44

activated through a control button integrated into the simulation model. According to this order, an extraction vehicle serving as an emergency transport unit selects the safest available route to reach the affected area. Simultaneously, residents without vehicles begin walking toward the designated rendezvous point, where they wait for the extraction bus. Upon arrival, evacuees board the bus, which then departs toward the shelter. However, instead of taking the main road, which becomes hazardous as the fire approaches, the vehicle dynamically reroutes through an alternative, longer but safer path to reach the shelter without exposure to the advancing flames.

4.3. Comparison of the results

The comparison between the evacuation drill and the simulation results indicates overall consistency in the evacuation process, with differences in specific temporal components. In the simulation, the preparation and walking time for self-evacuation was 25.8 min (21.8 + 4.0), while in the drill, the total time from alert to arrival at the rendezvous point (representing the preparation and walk time) was 22 min, representing a 3.8-minute difference. In the extraction scenario, the simulation recorded 23.9 min (20.56 + 3.33), compared with 22 min in the drill, indicating a 1.9-minute difference in the sum of preparation and walking time. These values are relatively close, and the deviation remains within a reasonable range, which may be attributable to variability in walking speeds and preparation durations across residents and cases.

The waiting time for self-evacuation in the simulation was 0.93 min, representing the time required to enter their car. For extraction, the simulation recorded a waiting time of 30.61 min, compared to 18 min during the drill. In the self-evacuation scenario, the short waiting time

Table 6

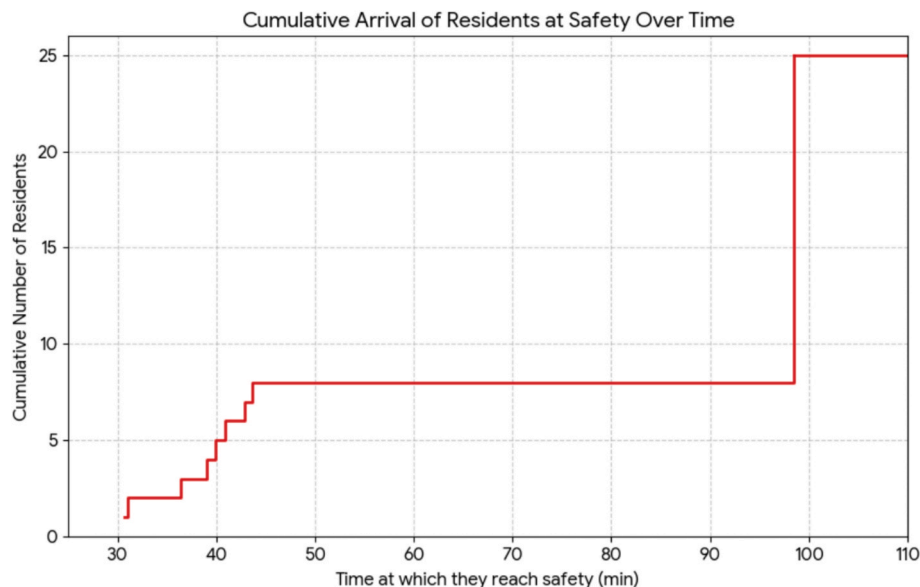
Minimum distance to fire of self-evacuated residents.

Car number (in the order of evacuation)	Minimum distance to fire (m)
1st	1037.58
2nd	836.88
3rd	692.99
4th	563.67
5th	468.35
6th	466.13
7th	290.01
8th	161.29
Extraction vehicle	1296.63

Table 7

Extraction vehicle performance indicators.

Parameter	Value
Dispatch time (min)	9.52
Time to reach the village (min)	43
Loading time (min)	3.4
Time to safety (min)	44
Total operation time (min)	100.9

**Fig. 5.** Number of evacuees reaching safety over time.

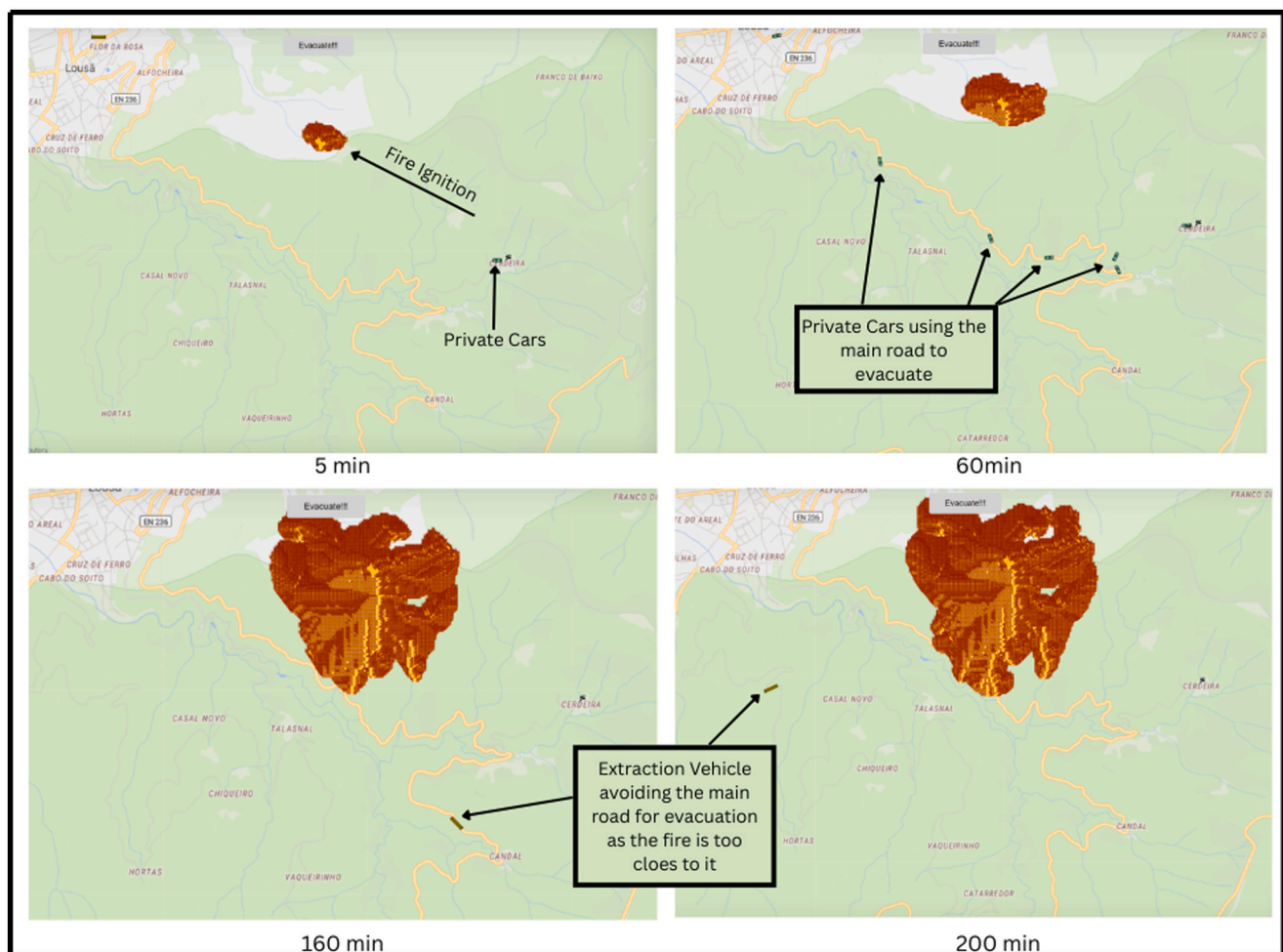


Fig. 6. Model at run-time.

indicates a lack of coordination needs, as residents immediately left in their private vehicles. Conversely, the longer waiting time in the extraction scenario is because, in the simulation, this period includes not only the coordination and loading of passengers but also the time for emergency vehicles to arrive at the village and the dispatch time specified in the simulation, which were not accounted for in the drill. If the bus dispatch time is excluded, the waiting time drops to 21 min, showing a 3-minute difference from the drill.

Regarding travel time to shelter, self-evacuating residents reached safety in 11.3 min in the simulation, using a direct route that was shorter and faster but exposed to a higher risk due to fire propagation along the main road. For the extraction scenario, the simulated travel time was 44 min, while the drill recorded 50 min, corresponding to a 6-minute difference. The difference between the drill and the simulation in this phase primarily results from differences in travel speeds: the model employs a higher average velocity than that observed during the actual exercise (i. e., a lower convoy velocity in the drill due to ethical and safety measures).

5. Discussion

As wildfires become more frequent and severe, their impacts on human settlements and ecosystems intensify, causing significant losses of life, property, and natural resources. While evacuation cannot prevent the destruction caused by the fire itself, it remains the most effective measure to safeguard human life when properly planned and executed. Evacuation simulation models are essential tools for designing and

testing these plans, enabling authorities to anticipate challenges, assess strategies, and adapt responses to the specific characteristics and needs of each community (Ronchi et al., 2019; Beyki et al., 2023; Lovreglio et al., 2020).

The model presented in this work is not intended to perfectly replicate real-world behavior. Wildfires are complex physical phenomena influenced by countless interacting factors, which even advanced fire models cannot fully capture, and efforts are aiming to improve the wildfire simulation accuracy (Beyki et al., 2025a,b). When human behavior under risk is incorporated into this system, uncertainty increases. As George Box stated, “All models are wrong, but some are useful.” (Box, 1976), and our model is no exception. It offers a practical and adaptable framework to explore evacuation processes under wild-fire threat, rather than an exact reproduction of reality.

The model integrates the three main layers of the evacuation process, fire propagation, pedestrian movement, and vehicle traffic into a single, dynamically coupled environment. It directly addresses key gaps identified in the literature, including: (i) the lack of dynamic rerouting mechanisms that adapt to fire-induced road closures rather than destination omission, (ii) limited consideration of multimodal evacuation, (iii) inadequate coupling of high-resolution fire spread with agent-level decision-making, and (iv) insufficient representation of inbound evacuation and extraction operations. These improvements enable the model to simulate both self-evacuation and organized extraction scenarios, capturing realistic interactions among evacuees, vehicles, and advancing fires.

The results from the case study demonstrate the model’s feasibility

and its ability to reproduce realistic evacuation dynamics. The simulated evacuation closely matched the timings observed during the real-world drill, validating the consistency of the model's behavioral and operational components. This alignment supports its applicability for both research and operational use. Future work will include testing the model on real wildfire events to evaluate its performance under more complex, time-critical conditions.

The model is proposed for two main applications. The first is strategic planning, where emergency managers can use it to develop evacuation strategies and protocols without time pressure. In this mode, the user can run multiple scenarios with different input parameters, such as varying self-evacuation rates, compliance levels, or walking speeds, and analyze their impact on evacuation time and exposure risk. The model allows for numerous simulations with different values for one or more parameters or variables to be run simultaneously, with the results compared. This approach enables sensitivity testing and strategy optimization before an actual event occurs.

The second application is real-time incident management. In this context, time becomes critical. The simulation must run efficiently and rely on quickly accessible data. The model setup for a specific area of interest can generally be completed within 30 min, including the acquisition of essential inputs such as fuel maps, topography, meteorological data, road networks, and population distribution, all of which are freely available for most regions worldwide. The case file for each area can be saved and made ready to run simulations, saving time in the event of an incident. Authorities can have files prepared for their respective jurisdictions. The simulation itself can be executed in under five minutes on a standard workstation. For instance, the case study simulation ran on a computer with a 3.4 GHz processor and 32 GB of RAM in about one minute. This level of efficiency supports the model's potential use in operational decision-making during unfolding emergencies.

Finally, the reliability of any model-based decision depends on how well its assumptions and inputs reflect reality. Evacuation plans and decisions derived from simulation outputs must therefore be verified, tested, and revised before being applied in real situations. Effective planning and rapid, well-coordinated responses can significantly reduce fatalities and improve community safety. The proposed model serves as a step toward integrating scientific modeling and operational practice, providing a flexible and accessible tool for understanding, testing, and managing evacuation under the growing threat of wildfires.

This work has limitations that should be acknowledged. First, while the model integrates wildfire spread, pedestrian, and vehicle dynamics in a unified framework, each layer remains a simplification of the real-world processes it represents. The fire spread component, although based on high-resolution wind–fire coupling, relies on simulated outputs from FireStation and therefore does not allow full real-time feedback between fire behavior, suppression efforts, and evacuation actions. This limits the capacity to capture mutual interactions, such as the influence of human-made firebreaks on hazard progression during the evacuation simulation.

Second, human behavior is represented through parameterized probabilities and stochastic distributions rather than empirically derived behavioral data. Factors such as risk perception, decision timing, preparation time, and compliance with evacuation orders are modeled probabilistically, which means that the variability of real human responses cannot be fully captured. The measurement and acquisition of such data from real wildfire incidents remain a major challenge in evacuation simulation under wildfire threat, and extensive research is required to address this gap. More data from actual wildfire evacuation cases are required to better reflect human behavior under threat in models.

Third, the model assumes a static communication structure and perfect information transfer between agents and authorities. In practice, communication delays, misinformation, or technological failures can critically affect evacuation outcomes. Another limitation concerns the case study's scale and calibration. The model was tested on a small

settlement with a controlled population and limited network complexity. While this was suitable for validation, larger-scale or more urbanized WUI areas may require higher computational power and additional calibration to maintain runtime efficiency and accuracy.

Finally, the validation process was based on a single evacuation drill under ideal conditions. Real wildfire events often involve uncertainty, infrastructure degradation, and unpredictable environmental changes that are difficult to replicate in controlled exercises. Therefore, while the model demonstrates good agreement with observed data, further validation with empirical data from real incidents is necessary to fully establish its predictive reliability.

Future work will focus on expanding the model's capabilities and improving its realism and applicability across different contexts. One key direction is to strengthen the behavioral component by incorporating data from real evacuation events, surveys, or experiments to better capture human decision-making, hesitation, and compliance under stress. This would allow the model to move beyond probabilistic behavior and reflect more realistic individual and collective responses.

The model should also be tested in larger, more complex WUI areas to evaluate its scalability and performance under high population density and extensive transportation networks, as well as to conduct sensitivity analyses of the inputs. Such applications would help assess its ability to support regional evacuation planning and coordination among multiple agencies.

The current implementation does not incorporate density-dependent traffic dynamics. Because link travel times are parameterized and independent of traffic volume, the model does not simulate congestion, queue propagation, or intersection delays. As a result, evacuation clearance times may be underestimated in scenarios involving large populations or constrained urban networks where traffic interactions become a dominant factor. Future work will integrate congestion-aware traffic modeling, either through mesoscopic speed–density formulations or microscopic car-following approaches in the future model updates, to enhance applicability to larger-scale or high-density wildland–urban interface settings.

Additionally, incorporating communication networks, warning dissemination systems, and infrastructure disruptions would enhance the model's capacity to represent realistic operational conditions. Finally, validation with data from real-world wildfire evacuations will be essential to confirm the model's robustness and refine its predictive accuracy. These future developments will help transition the framework from a research-oriented tool to an operational platform that supports decision-making in both planning and emergency management.

6. Conclusion

The increasing frequency and severity of wildfires pose a growing threat to communities located in the wildland–urban interface, where traditional response measures often prove insufficient to protect human life. This work presents an integrated agent-based model that simulates the evacuation process under wildfire threat, combining fire spread dynamics, pedestrian movement, and vehicular traffic within a single adaptive framework. By addressing several of the key gaps identified in previous research, such as the lack of dynamic rerouting, limited treatment of multimodal evacuation, and the absence of inbound extraction operations, the model provides a more complete representation of how people and emergency services interact during a wildfire evacuation.

The application of the model to a real case study demonstrated its capacity to reproduce observed evacuation patterns and timings with reasonable accuracy, confirming the feasibility of the approach. The results highlight its potential to serve as both a planning tool for developing evacuation strategies and a real-time decision-support system during ongoing emergencies. Its modular structure, reliance on freely available input data, and fast computation make it accessible for a wide range of communities and authorities.

Although the model simplifies several aspects of human behavior

and hazard dynamics, it represents a significant step toward integrating detailed fire simulations with realistic evacuation processes. The framework can support emergency planners and researchers in evaluating different strategies, testing “what-if” scenarios, and improving preparedness for wildfire emergencies.

Future developments should focus on enhancing the model's behavioral realism, integrating real-time data streams from active fires, and extending its application to larger and more complex urban areas. Continued validation using empirical data from actual wildfire events will further strengthen its reliability. Ultimately, the model aims to support the design of safer evacuation protocols, contribute to evidence-based decision-making, and improve community resilience against the escalating threat of wildfires.

CRediT authorship contribution statement

Shahab Mohammad Beyki: Writing – original draft, Visualization, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Anne S. Patricio:** Writing – review & editing, Software. **António Gameiro Lopes:** . **Aldina Santiago:** Writing – review & editing, Supervision. **Luís Laím:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work is financed by national funds through FCT – Foundation for Science and Technology through the research project PCIF/AGT/0061/2019 (EVACUARFLORESTA) and the grant agreement [DPA–22–140] attributed to Shahab Mohammad Beyki.

This work was also partly financed by FCT/MCTES through national funds (PIDDAC) under the R&D Unit Institute for Sustainability and Innovation in Structural Engineering (ISISE), under reference UIDB / 04029/2020 (doi.org/10.54499/UIDB/04029/2020), and under the Associate Laboratory Advanced Production and Intelligent Systems ARISE under reference LA/P/0112/2020.

This work is financed by national funds through FCT – Foundation for Science and Technology, under a grant agreement [2023.00658.BD] attributed to the first author.

Data availability

Data will be made available on request.

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